

ANIMAL BREEDING

Last five-year previous year question 2019-2023

1. _____ refers to the mating of more closely related individuals within the same breed for 4- 6 generations
- a. Inbreeding
 - b. Outcrossing
 - c. Out breeding
 - d. Interspecific hybridisation
2. This is the practice of mating of animals within the same breed, but having no common ancestors on either side of their pedigree up to 4-6 generations is called as _____.
- a. Outbreeding
 - b. Outcrossing**
 - c. Inbreeding
 - d. Interspecific hybridisation

Answers – 1. A 2.B

Mains 2021

1. What is cross breeding? what are the advantages and disadvantages of cross breeding and also write its application in animal husbandry? (10 Marks)

ANIMAL BREEDING

Animal breeding is the practice of **controlling the reproduction of animals to produce offspring with specific traits**. This has been done for thousands of years, with farmers selectively choosing animals with desirable qualities to breed together.

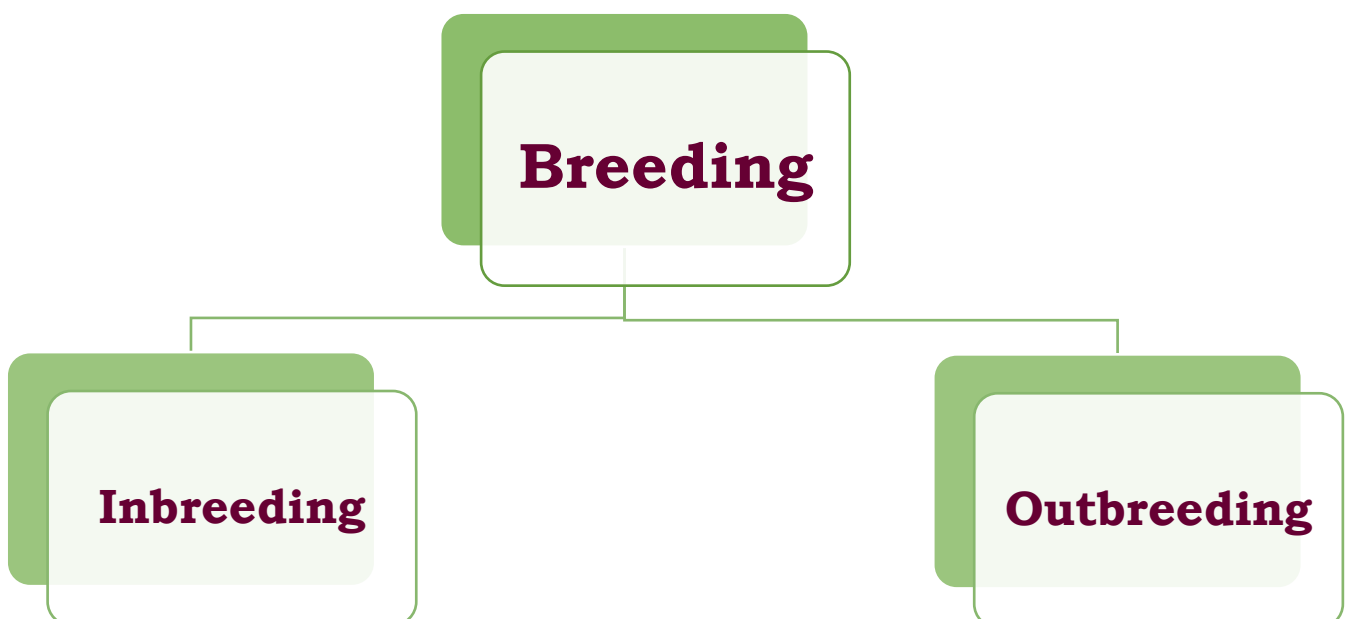
Animal breeding is a big part of agriculture and **has helped to increase food production around the world**. It can also be used to improve the health and well-being of animals

Animal breeding aims at increasing the yield of animals and improving the desirable qualities of the produce.

Mating of good quality animals to produce highly productive and suitable animals for enhancement of overall performance in the subsequent generations and to augment production and profitability is termed animal breeding.

BREED - A group of animals related **by descent and similar in most characters like general appearance, features, size, configuration, etc.**, are said to belong to a breed.

Methods Of Animal Breeding



1. INBREEDING

Inbreeding:

Inbreeding refers to **the mating of more closely related individuals within the same breed for 4- 6 generations**. The breeding strategy is **as follows – superior males and superior females of the same breed are identified and mated in pairs**.

The progeny obtained from such mating's are evaluated and superior males and females among them are identified for further mating. **A superior female, in the case of cattle, is the cow or buffalo that produces more milk per lactation**. On the other hand, a superior male is the bull, **which gives rise to superior progeny** as compared to those of other males.

Advantages Of Inbreeding

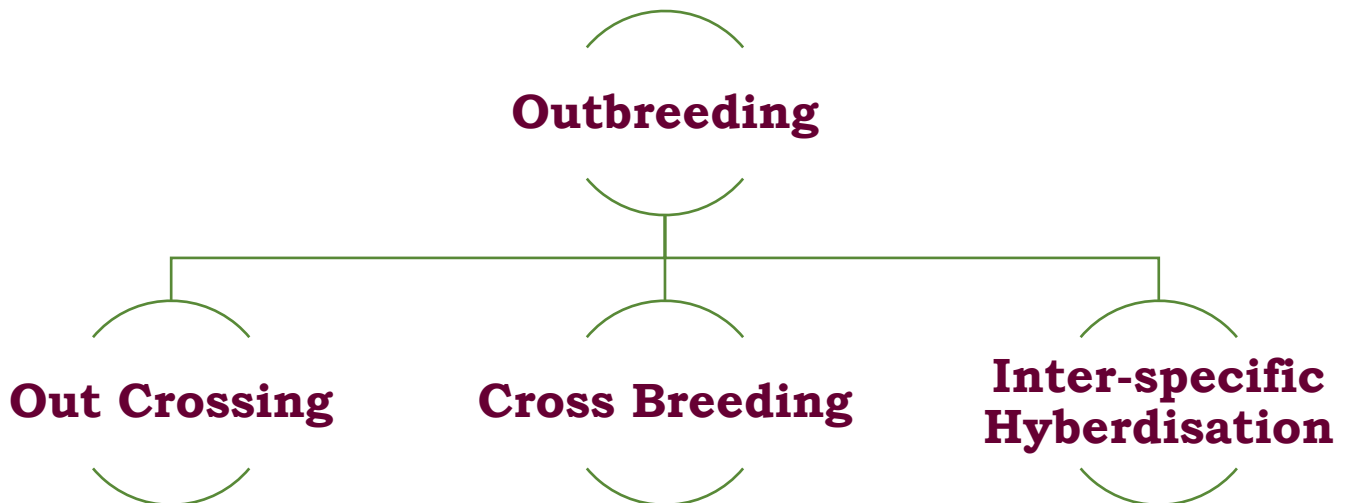
- **Increased homozygosity for desirable traits:** Inbreeding can concentrate desired genes from a particular ancestor. **If an animal possesses a highly valuable trait, breeders can use inbreeding to create offspring more likely to inherit that trait.** This can be **helpful in quickly establishing a new breed with specific characteristics**.
- **Exposure and elimination of recessive genes:** Inbreeding can **bring hidden recessive genes, both good and bad, to light**. Breeders can **identify and eliminate undesirable recessive traits** by observing their increased presence in inbred offspring. This allows for selection against these traits over time. It also **helps in accumulation of superior genes and elimination of less desirable genes**. Therefore, this approach, where there is selection at each step, increases the productivity of inbred population
- **Predictable breeding lines:** Highly inbred **animals often have a more predictable genetic makeup**. This can be advantageous for breeders who want to know exactly what traits offspring will inherit from specific parents.
- **Inbreeding is necessary** if we want to evolve a pure line in any animal.

Disadvantage

- **Continued inbreeding, especially close inbreeding**, usually reduces fertility and even productivity
- **Inbreeding depression** – inbreeding depression develops due to continues breeding of similar breeds. **Breeders have long known that inbreeding usually leads to a reduction in fitness, owing to the deterioration in important attributes**, such as fertility, vigour and resistance to disease. This is known as **inbreeding depression which** Reduced genetic diversity
- **Emergence of some non-desirable characters** in the breed like dwarfism, reduced milk production etc.

2. Out-Breeding:

Out-breeding **is the breeding of the unrelated animals**, which may be between individuals of the same **breed but having no common ancestors for 4-6 generations** (out-crossing) or between different breeds (**cross-breeding**) or different species (**inter-specific hybridisation**).



a. Out-Crossing-

This is the practice of mating of **animals within the same breed, but having no common ancestors on either side of their pedigree up to 4-6 generations**. The offspring of such a mating is known as an out-cross. It is the **best breeding method for animals** that are

below average in productivity in milk production, growth rate in beef cattle, etc. A single outcross often helps to overcome inbreeding depression

b. Cross-Breeding:

In this method, **superior males of one breed are mated with superior females of another breed. Cross-breeding allows the desirable qualities of two different breeds to be combined.** The progeny hybrid animals may themselves be used for commercial production. Alternatively, they may be subjected to some form of inbreeding and selection to develop new stable breeds that may be superior to the existing breeds. Many new animal breeds have been developed by this approach. **Hisardale is a new breed of sheep developed in Punjab by crossing Bikaneri ewes and Marino rams.**

c. Interspecific hybridisation:

In this method, **male and female animals of two different related species are mated.** In some cases, the progeny may combine desirable features of both the parents, and may be of considerable economic value, **e.g., the mule (a cross between donkey and horse)**

Type of Outbreeding	Mating Individuals	Purpose
Outcrossing	➤ Same breed, ➤ no common ancestors for several generations	➤ Improve genetic diversity, ➤ Reduce inbreeding depression, ➤ maintain breed characteristics
Crossbreeding	➤ Different breeds of the same species	➤ Enhance hybrid vigor, ➤ combine desirable traits, ➤ create new breeds
Interspecific hybridization	➤ Different species	➤ Introduce entirely new traits, ➤ study genetic compatibility

Advantages Of Out Breeding

Outbreeding, the breeding of individuals from genetically distinct populations, offers a number of advantages over inbreeding

- **Increased genetic diversity:** Outbreeding introduces *new alleles (versions of genes) into the gene pool, creating a more diverse population.* This diversity provides a broader range of potential traits and adaptations for offspring.
- **Reduced risk of inbreeding depression:** As discussed earlier, inbreeding depression can lead to a decline in health and fitness. **Outbreeding helps to avoid this by preventing the concentration of harmful recessive alleles.**
- **Heterosis (hybrid vigor):** This is a phenomenon where offspring from outbreeding often exhibit increased fitness and vigor compared to their parents. **They may grow faster, have higher fertility, be more resistant to diseases,** and have a stronger immune system.
- **Improved adaptability:** A more diverse **gene pool allows a population to better adapt to changing environmental conditions.** This can be crucial for survival in the face of disease outbreaks, climate change, or other challenges.
- **Reduced expression of deleterious genes:** **Recessive alleles for harmful traits are less likely to be expressed in offspring** when masked by dominant alleles from an unrelated parent.
- **Creation of new breeds:** Outbreeding **different breeds or populations** can lead to the development of entirely new breeds with unique and desirable combinations of traits.
- **Growth rate and productivity:** Animals **may grow faster, produce more milk or eggs,** or have higher meat yields.

- **Disease resistance:** Outcrossed offspring **can have a broader immune system due to the wider range of genes inherited**, making them more resistant to a variety of diseases.
- **Fertility and reproductive health:** Outcrossing **can improve fertility rates** and lead to healthier offspring.
- **Complementing breed weaknesses:** Outcrossing allows breeders to address weaknesses within a particular breed. For example, **outcrossing a breed known for high milk production but lower disease resistance with a breed known for strong disease resistance** can create offspring with both desired traits.

Disadvantages

Outbreeding, while generally beneficial, does come with some potential drawbacks.

Here are some disadvantages to consider:

- **Loss of desired traits:** When breeding with **very distantly related individuals**, **there's a chance of losing some of the desirable traits** that were established through previous breeding efforts. This can be a concern for breeders who are trying to maintain specific characteristics in a population.
- **Outbreeding depression (in rare cases):** While uncommon, outbreeding **depression can occur in situations where two very different populations are crossed**. This can happen if the parental populations are adapted to very different environments or have significant **genetic incompatibilities**. The offspring may be infertile, have developmental issues, or lack the vigor typically associated with outbreeding.
- **Disruption of co-adapted gene complexes:** Over time, natural selection **can favor combinations of genes that work well together**. Outbreeding can disrupt these co-adapted gene complexes, potentially leading to a decrease in overall fitness in the offspring until new beneficial combinations arise.

- **Acclimation issues:** If breeding animals **from vastly different environments, the offspring may not be well-suited to either parent's environment.** They might lack the necessary adaptations to thrive in either location.
- **Unpredictable results:** Outbreeding introduces **more genetic variation, which can lead to more unpredictable outcomes in terms of offspring traits.** While this can be positive for creating new variations, it can also make it harder for breeders to achieve specific goals.

Application Of Out Breeding

Outbreeding practices are applied in various ways across different fields, particularly in agriculture and conservation efforts. Here are some key applications:

Agriculture

1. **Improved livestock production:** Outbreeding is a **cornerstone of modern livestock breeding.**

Breeders use outcrossing and crossbreeding to achieve:

- **Increased milk production** in cows ex- Frieswal a cross of HF and HF×Sahiwal
- **Faster growth rate in chickens and pigs**
- **Improved disease resistance** in various animals
- **Enhanced meat quality**

2. **Maintaining Genetic Diversity:**

Outbreeding programs help to **maintain genetic diversity within breeds, which is crucial for long-term population health and adaptability.**

3. **Creation Of New Breeds:**

Crossbreeding has been **instrumental in developing entirely new breeds of animals with specific traits suited for different purposes and environments.** Ex mule, dolly sheep etc

Controlled Breeding –

Controlled breeding experiments are carried out using artificial insemination. The semen is collected from the male that is chosen as a **parent and injected into the reproductive tract of the selected female by the breeder.** The semen may be used **immediately or can be frozen and used at a later date.** It can also be transported in a frozen form to where the female is housed. In this way desirable matings are carried. **Artificial insemination helps us overcome several problems of normal matings**

Multiple Ovulation Embryo Transfer Technology (MOET)-

It is one such programme for herd improvement. In this method, **a cow is administered hormones, with FSH-like activity, to induce follicular maturation and super ovulation – instead of one egg, which they normally yield per cycle, they produce 6-8 eggs.** The animal is **either mated with an elite bull or artificially inseminated.** The fertilised eggs at 8–32 cells stages, are recovered non-surgically and transferred to surrogate mothers. The genetic mother is available for another round of super ovulation. **This technology has been demonstrated for cattle, sheep, rabbits, buffaloes, mares, etc. High milk-yielding breeds of females and high quality** (lean meat with less lipid) **meat-yielding bulls have been bred successfully to increase herd size in a short time.** the success rate **of crossing mature male and female animals is fairly low even though artificial insemination is carried out. To improve chances of successful production of hybrids MOET are used.**